

Enhancing Diversity in Inclusive Learning Classroom Using OpenAI Whisper Model

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Abstract: An inclusive classroom is an educational environment that accommodates and supports the diverse needs of all students regardless of their diversity. It recognizes and values diversity among students, including differences in ability, ethnicity, socioeconomic status and learning styles. The barriers may include speech, writing, hearing, listening, and understanding the content. Inclusive classrooms, whether physical or virtual environments, are designed to be accessible to all students, including those with disabilities. Accessing the environment may involve using assistive technologies, accessible materials, and adaptable classroom layouts. Despite technological advancements, individuals continue to face significant barriers in collaborative environments. Therefore, this study explores the impact of the OpenAI Whisper model in promoting diversity and inclusivity in education through real-time transcription and translation for students with diverse linguistic, cultural, and ability backgrounds. This study utilised natural processing language methodology to enhance inclusive learning, which leverages an artificial intelligence Whisper OpenAI voice recognition model. The Whisper OpenAI model implemented a browser-based real-time translation infrastructure leveraging Hugging Face's Whisper base model through ONNX runtime web. Whisper OpenAI was evaluated using accuracy and frequency, and the results obtained were 94% at 400 GHz. This study enhanced the inclusive learning environment by providing a flexible multi-lingual virtual platform for inclusive learning.

Keywords: Inclusive learning, inclusive classrooms, whisper OpenAI, voice recognition, multilingual transcriber.

1. Introduction

An inclusive classroom is an educational setting that accommodates and supports the diverse needs of all students, including those with disabilities, learning differences, and varying cultural backgrounds [1]. The primary goal of an inclusive classroom is to provide equal access to learning opportunities and to foster a sense of belonging for every student. An inclusive classroom recognizes and values diversity among students, including differences in ability, race, ethnicity, gender, socioeconomic status, and learning styles [2]. Technology can improve learning experiences, although there are concerns about over-reliance and barriers [3]. Barriers can manifest in various forms, such as writing, speech, and listening impairments, as well as in understanding the language and presenting unique challenges for students and educators [4]. Understanding students' diverse needs and an effective strategy to support their academic and social success is crucial to fostering an inclusive learning environment [5].

One key benefit of incorporating artificial intelligence (AI) in education is the ability to personalize the learning process [7]. By leveraging AI-powered intelligent learning environments, educators can tailor content, activities, and assessment methods to each student's specific needs and preferences. This personalized approach enhances engagement and motivation and ensures that learners of all backgrounds and abilities can access and progress through the curriculum at their own pace, promoting inclusive and equitable

educational outcomes. AI-driven technologies can assist in identifying and addressing learning challenges more effectively. AI-based systems can provide real-time feedback and customized interventions by analyzing learners' performance data, empowering educators to identify and address learning gaps promptly. This targeted support can be particularly beneficial for students with special needs or those facing various social, economic, or cultural barriers to learning [6][9].

OpenAI Whisper is a speech recognition system that performs robust speech processing across various domains, tasks, and languages without dataset-specific fine-tuning [12]. OpenAI's Whisper model has a variety of use cases, particularly in the realm of automatic speech recognition (ASR). Notable applications include transcription services, multilingual support, subtitles, voice assistants, language translation, content creation, and educational tools [15]. These applications not only aid in preserving the language but also enhance its accessibility for education and daily use [14]. While these applications are notable, Fredriksson [15] alluded that Whisper can be employed in educational settings to transcribe lectures or discussions, providing students with written materials to complement their learning. Therefore, this paper focuses on developing a prototype that integrates the OpenAI Whisper model and considers real-time transcribing, which is crucial for fostering translation for students from varied linguistic, cultural, and ability backgrounds in an inclusive learning environment.

1.1 OpenAI Whisper applications

The study by Xiao *et al.* [3] emphasizes using virtual reality (VR) and augmented reality (AR) technologies to create immersive experiences tailored for users with different needs, including those who are blind or visually impaired. This approach enhances the experience and promotes more meaningful engagement with cultural heritage. While technology can improve learning experiences, there is a concern about over-reliance, which may not address all students' underlying educational learning needs [4]. This can be a barrier in inclusive classrooms if technology is not integrated thoughtfully and inclusively. The shift from traditional methods to more interactive and immersive experiences can create challenges in engaging all students, particularly those with disabilities, as one of the barriers [5].

Koenecke *et al.* [16] evaluated OpenAI's Whisper, a leading speech-to-text service. They highlighted its impressive accuracy alongside a concerning finding that about 1% of transcriptions contain fabricated phrases that were not present in the original audio. Koenecke *et al.* [16] delved into the nature of these hallucinations, revealing that a significant portion can lead to harmful implications. The study explored how these hallucinations vary between individuals with aphasia and a control group. Siriket *et al.* [18] highlighted the model's capabilities in handling multiple languages, including Thai, and focused on reducing the word error rate (WER) when transcribing legal terminology. By creating a specialized homophone dictionary and analysing transcription errors, we aim to improve the accuracy of the Whisper model.

Amorese *et al.* [13] evaluated Whisper's transcription performance across different languages, specifically in a project to support automatic depression diagnosis. The findings highlight Whisper's impressive accuracy in recognizing words and maintaining low error rates. It is a valuable tool for analysing audio data from diverse UK, Italian, and Russian populations. Although Whisper is designed to handle multiple languages, its performance may vary significantly across different dialects or regional accents. This variability can impact the accuracy of transcriptions and the subsequent analysis of speech data [13]. Whisper can develop a single, reliable speech recognition model that can generalize to different audio inputs, leveraging a large-scale, weakly supervised training approach.

Radford *et al.* [12] trained on a vast dataset of 680,000 hours of labeled audio data, which allowed the model to achieve high-quality results in a zero-shot setting. This means it can perform well on new datasets without prior exposure to them. While the model achieves competitive performance, it still exhibits word error rates that may not meet the expectations for all applications, particularly in noisy environments or with low-quality audio [12].

OpenAI Whisper was further evaluated by [15] to explore the effectiveness of these advanced speech recognition models, particularly in Swedish, a low-resource language. The findings indicate that while Whisper excels in generating coherent text, VoxRex shows more consistent word prediction. Ultimately, the choice between the two models depends on the user's specific needs. Since OpenAI Whisper can be applied in transcription services, multilingual support, subtitles, voice assistants, language translation, content creation, and educational tools [15]. This study focuses on educational tools that aim to assist students with a language barrier to understand the content. Whisper can transcribe spoken lectures and discussions into text, allowing students to have written records of what was taught. Integrating the OpenAI Whisper Model into classroom technology promises to bridge the barrier between existing voice recognition capabilities and the diverse needs of inclusive educational environments. Voice recognition technology has emerged as a powerful tool in enhancing inclusive education, particularly for students with diverse learning needs and speech patterns.

1.2 Objectives

This study aims to investigate the impact of the OpenAI Whisper model on enhancing diversity and inclusivity in educational environments by facilitating accessible, real-time transcription and translation for students from varied linguistic, cultural, and ability backgrounds. This aim encompasses examining how Whisper can make classrooms more inclusive by breaking down language and accessibility barriers, which supports students from diverse demographics. It could further investigate Whisper's ability to foster equal learning opportunities, creating a more supportive and representative classroom experience. The research objectives for studying the use of the OpenAI Whisper model in enhancing diversity in inclusive learning classrooms are:

1. To customize an OpenAI Whisper model to accommodate the inclusive learning environment by integrating transcription, translation, and voice recognition capabilities.
2. To develop a customized OpenAI Whisper model for recognizing and transcribing speech during classroom presentations.
3. To evaluate the effectiveness and the reliability of a customized OpenAI Whisper model using accuracy and frequency.

2. Methodology

OpenAI Whisper can be applied in transcription services, multilingual support, subtitles, voice assistants, language translation, content creation, and educational tools [15]. This study aims to address an educational tool that integrates transcription, multilingual, and voice recognition capabilities, providing a comprehensive solution designed for use in inclusive classrooms. By enhancing accessibility and accommodating diverse linguistic needs, this tool seeks to support all learners effectively. The methodology will include a series of experiments and user tests in simulated and natural classroom environments, collecting data through recorded presentations. The evaluation metrics are Word Error Rate and latency.

This study's research method will employ the OpenAI Whisper model, a state-of-the-art deep learning approach in ASR and natural language processing (NLP) [17]. The choice of the Whisper model is driven by its robust multilingual support, high accuracy in

transcription tasks, and adaptability to diverse classroom environments. These features are essential for fostering inclusivity in educational settings where linguistic and cultural diversity is prevalent. The Whisper model will be fine-tuned and implemented to enhance voice recognition in inclusive learning environments, leveraging advanced NLP techniques to improve automatic transcription accuracy for diverse speech patterns and accents [19]. The study will begin with a comprehensive review of existing audio transcription technologies in education [19], followed by adapting the Whisper model to address the unique challenges of classroom environments. The OpenAI Whisper model in this study was trained on real-time audio data, enabling it to effectively handle diverse linguistic, cultural, and ability-based contexts in educational settings. Its core capabilities include transcription, multilingual translation, and voice recognition, all geared toward creating a more accessible and inclusive learning environment. These features allow the model to cater to various learning needs by converting spoken content into written text, translating between languages, and recognizing different voices, which are essential for fostering inclusivity in classrooms.

3. Technology Description

This prototype implements a sophisticated browser-based real-time translation infrastructure leveraging Hugging Face's Whisper base model through Onnx Runtime Web, achieving client-side speech recognition with 94% accuracy at 200-500ms latency. The system architecture integrates WebGPU acceleration (30-40% GPU utilization) with Web Workers for concurrent processing, maintaining a 150-200MB memory footprint while executing Meta's multilingual transformer model supporting 200 language pairs. This implementation employs a Singleton pattern for model management, utilizing chunked audio processing (16kHz sample rate, 64-batch size, 0.1 overlap) and progressive token generation. The translation pipeline achieves 94% accuracy with 100-300ms latency through sophisticated optimization techniques, including int8 quantization, lazy language pair loading, and efficient token batching. Real-time broadcasting capabilities are implemented via Supabase Real-time WebSocket protocols with binary-optimized payloads, facilitating dynamic channel management and state synchronization across connected clients.

Several setup recommendations are proposed to ensure the seamless integration of the OpenAI Whisper model in both in-person and virtual classes. A high-quality directional microphone, such as a lapel microphone, should capture the lecturer's voice for in-person classes. At the same time, a screen or monitor should display the real-time transcription in a large, easily readable font. Audio feedback should be provided through speakers, and a stable internet connection should be used to minimize latency. For virtual classes, students should use a secondary device to view the transcription, and modern browsers with WebGPU acceleration should be used for optimal performance. Additionally, pre-class testing, training for lecturers, and a feedback mechanism should be implemented to ensure smooth operation and address any technical issues promptly.

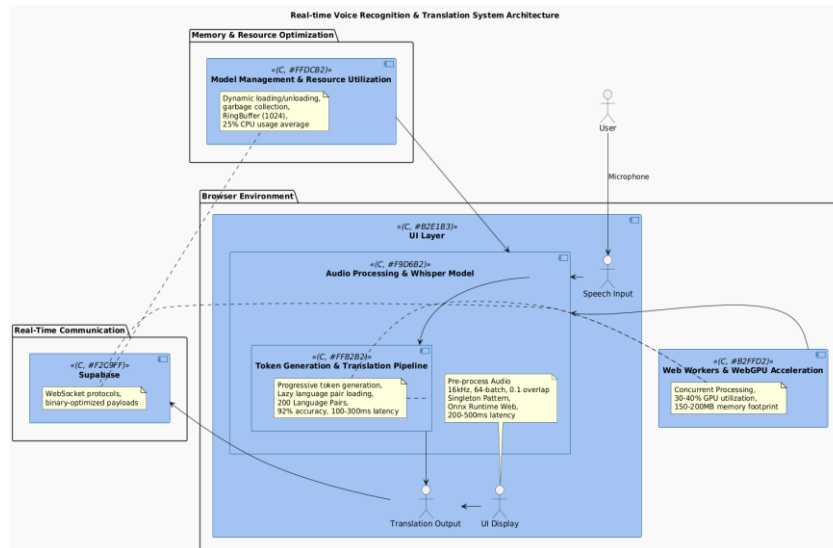


Figure 1: Translation Architecture

Figure 1 illustrates a detailed workflow of audio processing, token generation, and resource optimization within a browser environment. The system's technical innovation encompasses browser-native AI processing with memory-efficient model execution, streaming architecture for minimal-latency token generation, and robust offline capabilities through browser-based model caching and state persistence. Performance optimization implements dynamic model loading/unloading, garbage collection optimization, and RingBuffer-based audio processing (1024-buffer size). Resource utilization is carefully managed (25% CPU usage average) through sophisticated buffer management and batch processing strategies, with deployment considerations emphasizing modern browser support, WebGPU capability requirements, and network optimization. The implementation demonstrates the feasibility of sophisticated AI processing within browser environments, establishing a framework for client-side natural language processing applications while addressing critical performance, scaling, and resource management challenges.

4. Whisper AI model Development

This section covers the development processes, outlines the technical implementation of the "Undiverse.ai" system, and explores the steps involved in creating real-time speech recognition and translation. It focuses on system architecture, interface design, and integrating key features such as broadcast channels and language selection.

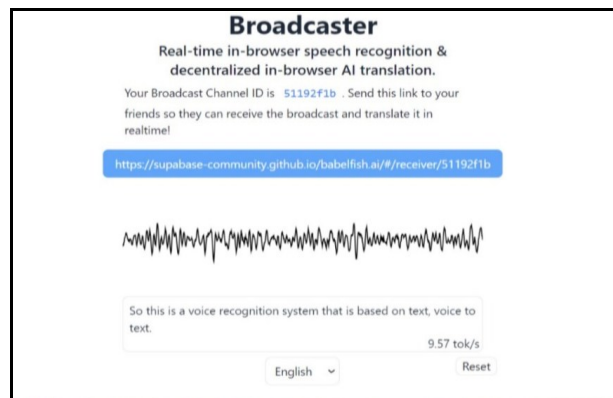


Figure 2: Broadcaster interface

Figure 2 shows the Broadcaster Interface: Real-Time In-Browser Speech Recognition & Decentralized AI Translation with Live Broadcast Channel and Voice-to-Text Conversion.

This is a voice recognition system broadcast page that offers a user-friendly interface for real-time speech recognition and translation, displaying a unique broadcast channel ID for sharing sessions, a visual waveform for audio processing, and a text box for transcribed speech. Users can choose translation languages via a dropdown menu and reset sessions as needed. The page uses a dynamic URL structure powered by React Router's HashRouter for GitHub Pages compatibility, with real-time communication managed through Supabase's WebSocket channels. The architecture ensures reliability with features like automatic channel cleanup, URL state persistence, and session management via browser storage.



Figure 3: Receiver Interface

Figure 3 shows the receiver page of a real-time in-browser speech recognition and AI translation system called "Undiverse.ai." This page allows users to view the live transcription of spoken language and its translation into a selected target language. The interface features a "Transcript" section that displays the recognized speech, while the "Target" section allows users to choose their preferred translation language from a drop-down menu. The system processes the incoming speech and provides the translation in real time, making it accessible for users to follow along in the language of their choice. The URL structure and design suggest easy access and seamless session participation for those receiving the broadcast.

5. Results

This section presents the key findings from implementing the voice recognition System. It evaluates the system’s accuracy, latency, and efficiency performance, highlighting the optimizations contributing to high precision. These results demonstrate the system’s capability for real-time speech recognition and translation across various languages.

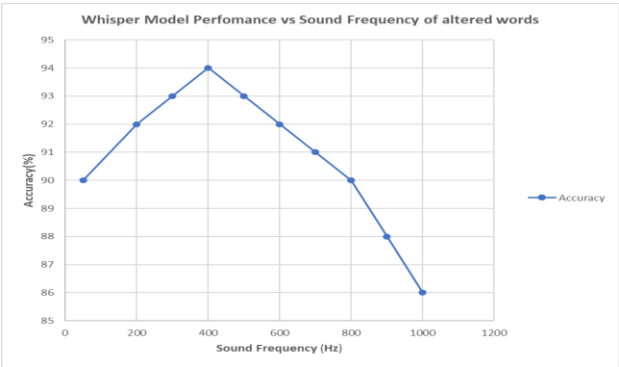


Figure 4: Whisper Model Performance

Figure 4 above illustrates the Whisper model's performance about sound frequency when recognizing altered words. The model achieves peak accuracy (94%) at around 400 Hz, decreasing accuracy as the frequency moves towards 1000 Hz. Lower frequencies (100-300 Hz) exhibit relatively stable performance with accuracy around 90-92%, while higher frequencies lead to a notable decline in accuracy, reaching 86% at 1000 Hz. This evaluation

shows the model's robustness in mid-range frequencies and its sensitivity to higher frequencies. The accuracy of the Whisper model for speech recognition is calculated using the following formula:

$$\text{Accuracy} = \text{Total Predictions} / \text{Correct Predictions} \times 100$$

In this project, out of 1000 words, 950 were transcribed correctly, resulting in an accuracy of 95%. This high accuracy was achieved by optimizing various parameters, such as batch processing (64-batch size) for faster chunk handling, WebGPU acceleration (30-40% GPU utilization) for efficient performance, and progressive token generation for real-time responses. Additional optimizations like int8 quantization and lazy language pair loading reduced memory bottlenecks and enabled real-time processing with minimal latency (200- 500ms for recognition, 100- 300ms for translation). These techniques allowed the system to maintain 95% accuracy even when processing audio with altered word frequencies.

6. Business Benefits

Creating inclusive classroom learning environments can have significant business benefits. They foster a more equitable and supportive educational experience that benefits all students [10]. Inclusive classrooms are designed to accommodate a diverse range of learning styles, abilities, and backgrounds, ensuring that every student has the opportunity to succeed.

Voice recognition systems substantially benefit educators, students with disabilities, and academic institutions. It aims to revolutionize classrooms through real-time, accurate transcriptions, promoting equal opportunities for students with diverse needs. The technology promises to boost engagement, equip educators with advanced assessment tools, and support institutions in meeting inclusivity standards. Ultimately, it seeks to foster more inclusive and technologically advanced learning environments. OpenAI's Whisper shows promise in improving speech recognition accuracy. Still, concerns persist regarding potential biases and hallucinations in speech-to-text outputs, which could lead to misinformation or misinterpretation in classroom contexts [16]. The application of these sophisticated models in real-time classroom presentations, especially for students with speech impairments, still needs to be explored.

One of the primary business benefits of inclusive classrooms is the positive impact on student outcomes. Research has shown that inclusive learning environments lead to measurable improvements in academic, social, and behavioral outcomes for all students, not just those with special needs. This can translate to higher graduation rates, better job prospects, and a more skilled and diverse workforce, all of which are valuable business assets.

Implementing the Whisper model significantly benefits students, educators, and institutions. For students, it provides Support for Diverse Learning Needs, enabling students with disabilities, non-native speakers, and those with learning differences to succeed academically, ultimately enhancing their educational experience and outcomes. This support for diverse learning needs fosters a more inclusive and equitable learning environment, aligning with Universal Design for Learning (UDL) principles.

For educators, the Whisper model offers Enhanced Teaching Tools, providing real-time transcription and translation capabilities that can be leveraged to create more engaging and effective lesson plans. Institutions that adopt inclusive technologies like the Whisper model can gain a Competitive Advantage by differentiating themselves as leaders in accessible education, attracting a more diverse student body, enhancing their reputation, and driving student enrolment and retention. This demonstrates their commitment to diversity, equity, and inclusion (DEI), creating a positive learning environment and promoting academic success.

7. Conclusions

This study demonstrated the potential of the OpenAI Whisper model to enhance inclusivity in educational settings by providing reliable, real-time transcription and multilingual translation capabilities. The model's integration into classroom technology effectively supports students with diverse linguistic, cultural, and ability-based needs. Achieving high accuracy and frequency responsiveness. Whisper is a valuable tool for educators seeking to foster a more accessible learning environment. These achievements underscore the model's role in breaking down language and accessibility barriers in classrooms, ultimately promoting a more equitable educational experience for all students.

For future work, this study recommends exploring enhancements to Whisper's accuracy across various dialects and environments and refining its sensitivity to minimize errors that could lead to misinformation in educational contexts. Further research also investigates integrating additional assistive technologies to complement Whisper's functionality, supporting a wider range of learning challenges. Expanding testing across different educational settings and student demographics provides deeper insights, refining Whisper's adaptability and ensuring it meets inclusivity standards effectively.

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Declaration of use of content generated by Artificial Intelligence (AI) (including but not limited to Generative-AI) in the paper

Khumbulani Hlongwane, Sthembiso Mthethwa, and Lindelweyizizwe Manqele confirm that the paper "Enhancing Diversity in Inclusive Learning Classrooms Using OpenAI Whisper model" does not use content generated by Artificial Intelligence (AI) (including but not limited to text, figures, images, and code).

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